

Murty-Simon at $n=30$ - verification companion

Fan-free reviewer edition 2; historical v1 preserved

Paul Lenz research project; mathematical development and drafting with ChatGPT/Geeps

9 September 2026

Abstract

This companion records the direct Fan-free upper-range replacement, the hostile assembly audit prompted by external critique, exact computational provenance, and preserved proof history. Fan's 1987 result remains cited as historical context but is not a logical dependency of the current project proof. Same-assistant checking is not external verification.

Claim under review. $e(G) \leq 225$, equality exactly $K(15,15)$.

Fan-free reduction. `project/research/fan-free-fixed-orders/2026-09-09-v1/FAN_FREE_REDUCTION.md`.

Hostile assembly audit. `project/research/fan-free-fixed-orders/2026-09-09-v1/FAN_FREE_AUDIT.md`.

Included source: `project/reviews/n30/2026-09-09-fan-free-v2/HISTORY.md`

n=30 proof history - Fan-free edition 2

Frozen edition-1 source: `project/reviews/n30/2026-09-09-candidate-v1/PROOF.md`

SHA-256 at build time: `6a03ea7ce46d8f7e26921fcb03e2bcf58005d628b9763e481b647d10768b`

Edition 2 removes Fan's theorem as a logical upper-bound dependency, cites Fan historically, makes the unordered-pair selection convention explicit, and incorporates the complete Fan-free upper-range evidence in `project/research/fan-free-fixed-orders/2026-09-09-v1/FAN_FREE_REDUCTION.md`. The frozen edition-1 proof and all old evidence remain unchanged.

Included source: `project/research/fan-free-fixed-orders/2026-09-09-v1/FAN_FREE_REDUCTION.md`

Fan-free upper-range reduction for the current fixed-order candidate frontier

9 September 2026. Research direction: Paul Lenz. Mathematical development and internal checking: ChatGPT/Geeps.

Status: candidate fixed-order proof component with exact finite arithmetic. Independent mathematical and computational review remains OPEN.

1. Purpose and historical relation to Fan

The original fixed-order candidate manuscripts at $n=25, 27, 28, 29, 30$ used G. Fan's 1987 theorem

G. Fan, *On diameter 2-critical graphs*, Discrete Mathematics 67 (1987), 235-240, DOI 10.1016/0012-365X(87)90174-9,

as a convenient published upper reduction. Fan's result is historically important and stronger than the order-specific calculations below, so it remains cited.

This note removes **Fan's theorem as a logical dependency** of every current project fixed-order candidate proof at $n=25, 27, 28, 29, 30$. It does **not** claim to reproduce or supersede Fan's all-order theorem. Instead, each order is closed directly above the conjectured extremal value using the project's already stated witness/quasi-edge/residual lemmas together with exact finite arithmetic.

The old Fan-based manuscripts, reviewer packages, workflows, certificates and hashes remain frozen and preserved as historical versions. Reviewer-facing v2 editions cite Fan for history/attribution while using this note as the logical upper-range reduction.

2. Common ingredients

No new graph-theoretic hypothesis is introduced merely to remove Fan. The replacement uses consequences already stated and audited within the project:

1. degree-sum bounds;
2. direct/two-step witness deficit bounds where available;
3. the complement/quasi-edge construction;
4. the convention choosing **exactly one selected cross-edge for each missing unordered pair in B**;
5. the residual ledger

$$\begin{aligned} e(C) + r &= L, \\ e(F) &= r + t, \\ \sum d_i &= 2(r + t), \\ \sum \rho_b &= r; \end{aligned}$$

6. residual activity $\rho_b \geq 1$ when $t > 0$;
7. selected-pair/source/supplement capacity inequalities;
8. charging and threshold-capacity inequalities;
9. the residual h-index bound;

10. exact Hall/source-capacity duals and, only where required, exact integer Farkas certificates for the stated relaxations.

The finite enumerations deliberately retain nongraphical numerical states. Therefore eliminating the relaxation is safe: every actual graph satisfying the written bridge is represented, while many impossible numerical states are intentionally retained.

2.1 Selection semantics made explicit

Selection is indexed by missing **unordered** B-pairs: one representative quasi-edge is designated for each such pair. Opposite orientations of the same B-pair can never both be selected. Hence the map from selected edges to unordered source/supplement pairs is injective by construction.

Likewise, whenever a residual-injection argument forces an A-B edge whose two endpoints both miss an A-vertex, that edge cannot be selected for a B-B missing pair: the unique exception of a selected edge belongs to B, so a selected edge must dominate every A-vertex. These are consequences of the construction, not additional assumptions.

3. n=25 without Fan

Let $m = e(G)$.

If $\Delta \leq 12$, degree sum gives $m \leq 150$. If $\Delta \geq 17$, the independent complement maximum-degree theorem already used in the frozen manuscript gives $m \leq 155$.

For $\Delta = 13$, the witness-deficit inequality

$$m \leq C(h, 2) + h(25-h) + o(o-1),$$

$$2h+o \leq 325-2m$$

is monotone stronger as m increases. The complete table at $m=157$ is already strictly below 157, hence excludes every $m \geq 157$ at $\Delta = 13$.

The frozen $n=25$ residual proof excludes $m=157$ for $\Delta = 14, 15, 16$. For every larger degree-sum-possible count, the exact Fan-free outer verifier gives:

Delta	edge counts scanned	outer states	survivors
14	158..175	128,666	0
15	158..187	88	0
16	158..200	0	0

Therefore, without Fan,

$$e(G) \leq 156,$$

and the frozen equality chain forces $K(12, 13)$.

4. n=27 without Fan

If $\Delta \leq 13$, degree sum gives $m \leq 175$. If $\Delta \geq 18$, the independent complement maximum-degree theorem gives $m \leq 181$.

For $\Delta = 14$, the witness bound

$$m \leq C(h, 2) + h(27-h) + o(o-1),$$

$$2h+o \leq 378-2m$$

is again monotone stronger with increasing m ; the complete $m=183$ table therefore excludes every $m \geq 183$.

The frozen $n=27$ proof excludes $m=183$ at $\Delta=15, 16, 17$. Above 183:

- $\Delta=16$, all $m=184 \dots 216$: 8,880 outer states, zero survivors;
- $\Delta=17$, all $m=184 \dots 229$: empty numerical domain;
- $\Delta=15$, all $m \geq 186$: zero outer survivors.

Only $\Delta=15, m=184, 185$ require the inherited residual-column stage:

```
m=184: 651 outer survivors -> 49,073 canonical columns
      -> 1,175,094 labelled orbit mass -> 0 final survivors;
m=185: 10 outer survivors -> 388 canonical columns
      -> 10,036 labelled orbit mass -> 0 final survivors.
```

Scan and check modes agree on all counts and ordered-column digests; the replay directly verifies the recorded strict subset inequalities without invoking max flow.

Hence, without Fan,

$e(G) \leq 182$,

and equality remains exactly $K(13, 14)$.

5. $n=28$ without Fan

The conjectured maximum is 196. The frozen direct-197 proof already excludes $m=197$ at every maximum degree.

For every $m \geq 198$:

- $\Delta \leq 14$ is impossible by degree sum;
- $\Delta=16$ is excluded by the same pointwise charging bound used at 197: the right side depends only on the degree band while the required left side $b+2t$ increases with m ;
- $\Delta=17, \dots, 26$ are excluded by the same residual h -index inequalities used at 197, again with a fixed right side and increasing left side;
- $\Delta=27$ is the universal-vertex/star case.

Thus only $\Delta=15$ needed fresh arithmetic.

5.1 $\Delta=15, m=198$ and 199

The strengthened preparation uses charging, residual activity, $\delta(C) \geq 1$, threshold capacity and exact source-capacity Hall dual lower bounds. It then exhaustively scans residual rows and applies exact row-level threshold capacity plus the corrected threshold/source-flow relaxation. Floating infeasibility alone is never accepted.

At $m=198$ ($t=3$):

```
charging-domain profiles: 6,859
retained demand profiles: 186
residual states:         649,380
row survivors:           418
row-threshold rejects:   418
final survivors:         0
```

At $m=199$ ($t=4$):

```

charging-domain profiles: 3,263
retained demand profiles:      22
residual states:              61,363
row survivors:                 21
row-threshold rejects:         21
final survivors:                0

```

No final LP/Farkas call was needed in either scope because exact row-threshold capacity already closed every retained row.

5.2 Delta=15, m=200 through 210

The exact early kernel closes the entire demand domain:

```

m=200 (t=5): 1,155 profiles = 1,070 threshold + 67 source-count + 18 exact
m=201 (t=6): 250 profiles, all threshold-rejected;
m=202 (t=7): 15 profiles, all threshold-rejected;
m=203..210: 0 charging-feasible profiles.

```

Therefore no graph has $m \geq 198$. Together with the frozen direct exclusion of 197,

$e(G) \leq 196$,

with the existing equality chain giving exactly $K(14, 14)$.

6. n=29 without Fan

The target is 210 and the frozen proof excludes $m=211$.

For $m \geq 212$:

- Delta ≤ 14 is impossible by degree sum;
- the Section 3 witness argument at Delta=15 is monotone stronger as m increases and therefore excludes every $m \geq 211$;
- the pointwise charging argument at Delta=17 only strengthens with increasing t ;
- the residual h-index argument excludes Delta=18, ..., 27 more strongly as m rises;
- Delta=28 is the star case.

Only Delta=16 required fresh arithmetic.

6.1 Delta=16, m=212

Here $a=12, b=16, t=4$. An audited wrapper removes only the frozen preparation/row scripts' historical CLI restriction to $t \in \{2, 3\}$; no mathematical formula is changed.

The complete result is:

```

charging-feasible demand profiles: 2,032
threshold rejects:                  1,706
source-count rejects:               258
exact early Hall-dual rejects:      65
open demand profiles:               3
residual numerical states:          19,630
row survivors:                      2
exact late Farkas rejections:       2
final survivors:                    0
exact certificate RHS values:       -499, -434

```

Every late certificate is rebuilt in the independently reconstructed corrected threshold model and verified by exact integer arithmetic.

6.2 Delta=16, m=213 and above

The early exact-acceptance kernel gives:

```
m=213 (t=5): 586 profiles = 576 threshold + 7 source-count + 3 exact Hall-
m=214 (t=6): 79 profiles, all threshold-rejected;
m=215 (t=7): 1 profile, threshold-rejected.
```

The generic outer scan has zero survivors at every $m=216 \dots 232$, the degree-sum ceiling for $\Delta=16$.

Thus no $m \geq 212$ graph exists, and with the frozen exclusion of 211:

$e(G) \leq 210$,

with equality exactly $K(14, 15)$.

7. n=30 without Fan

The target is 225 and the frozen proof excludes $m=226$ while characterizing equality at 225.

For $m \geq 227$, degree sum gives $\Delta \geq 16$; $\Delta=29$ is the star case. For $\Delta=18, \dots, 28$, the frozen charging domain is already empty at $m=225, 226$; for fixed degree band, increasing m increases t and hence the required left side $b+2t$, so those empty domains remain empty for every larger m .

It remains only to cover $\Delta=16, 17$ through their degree-sum ceilings.

7.1 Delta=16, m=227 through 240

For the near boundary, the strengthened trusted kernel gives:

```
m=227 (t=3): 32,031 demand profiles -> 977 retained;
              11,903,678 residual states -> 6,245 row survivors;
              all 6,245 exact row-threshold rejected; zero survivors.

m=228 (t=4): 19,260 demand profiles -> 285 retained;
              2,034,522 residual states -> 683 row survivors;
              all 683 exact row-threshold rejected; zero survivors.

m=229 (t=5): 10,037 demand profiles -> 59 retained;
              245,666 residual states -> 29 row survivors;
              all 29 exact row-threshold rejected; zero survivors.

m=230 (t=6): 4,299 demand profiles -> 3 retained;
              7,730 residual states -> 0 row survivors.
```

For $m=231 \dots 240$, the early exact-acceptance kernel closes the full demand domain:

```
m=231 (t=7): 1,360 profiles = 1,346 threshold + 12 source-count + 2 exact
m=232 (t=8): 238 profiles, all threshold-rejected;
m=233 (t=9): 9 profiles, all threshold-rejected;
m=234..240: 0 charging-feasible profiles.
```

Thus the complete degree-sum range $m=227 \dots 240$ at $\Delta=16$ is excluded.

7.2 Delta=17, m=227 through 255

The complete explicit upper-range scan was run rather than relying only on monotonicity:

```
m=227 (t=6): 15 charging-feasible profiles, all threshold-rejected;  
m=228..255: 0 charging-feasible profiles at every edge count.
```

Workflow run 34398181000 completed successfully and committed the durable checkpoint checkpoints_N30_D17_UPPER.json.

Hence every $m \geq 227$ is impossible. Combining this with the frozen exclusion of 226 and equality analysis at 225 gives, without Fan,

```
e(G) <= 225,  
with equality exactly K(15,15).
```

8. Computational provenance

n=25 and n=27

Primary upper-range scan:

```
run 34393554788  
head commit efe189ad911a3968e5b7d63a3de43a05c664bfb2
```

n=27 Delta15 residual-column closure:

```
run 34393710385  
head commit 7cc1971604efdf8fea5ec11bbc3b8826868c55ff
```

n=28

Generic discovery scan:

```
run 34394955761
```

Trusted near-boundary closure:

```
run 34396969425  
m=198 artifact SHA256 cf71339f83d49fc1a53bfaee66e73061bebbc8e9cbb7488c0655  
m=199 artifact SHA256 820703855b824b638ea0e36dc8b83d61da05df9515d7b7536b38
```

High-surplus early kernel:

```
run 34397027512  
artifact SHA256 aef829a157de26e2c92ca3ea131778e290026988c079142d37c00023e0
```

n=29

Generic upper discovery scan:

```
run 34394955761
```

Exact m=212 trusted-kernel closure:

```
run 34396452149 — SUCCESS  
checkpoint checkpoints_N29_D16_M212.json
```

m=213..215 early closure:

```
run 34396555260  
artifact SHA256 a2f94dd6a81941ba767c68a2d5dc89a14cda8dab01b4b96c5391e0369a
```


n=30

Delta16 m=227..230 trusted closure:

```
run 34397132809
```

all four mathematical jobs completed; three final Git writes raced and the

Delta16 m=231..240 early closure:

```
run 34396592020
```

```
artifact SHA256 b92efe1f4afe87ec1e3fa7d912713b9e84d98a07f4a953fba5efcd9df2
```

Delta17 complete upper range:

```
run 34398181000 — SUCCESS
```

```
checkpoint checkpoints_N30_D17_UPPER.json
```

All arithmetic exclusions accepted as proof events are exact integer/rational checks. Floating-point LP is used only to propose dual/Farkas multipliers where applicable.

9. Trust boundary and status

Fan's 1987 theorem is now **historical attribution only** for the current fixed-order candidate frontier $n=25, 27, 28, 29, 30$; none of those candidate proofs requires Fan's bound logically.

This does not remove other published dependencies explicitly retained by the manuscripts, such as complement/total-domination or dominating-edge results where used. It also does not convert same-assistant computation into external review. The graph-to-residual/quasi-edge lemmas remain the principal mathematical trust boundary.

The old Fan-based manuscripts, reviewer packages, evidence, failed exploratory routes, workflows and hashes remain preserved. New reviewer-facing v2 editions must point back to those versions rather than overwrite them.

Included source: `project/research/fan-free-fixed-orders/2026-09-09-v1/FAN_FREE_AUDIT.md`

Hostile assembly audit — Fan-free fixed-order frontier

9 September 2026. **PASS internally; independent external review remains OPEN.**

This audit was written after an external-AI critique challenged two selection/residual semantics and the reliance on Fan's 1987 upper bound. The two semantic objections were resolved by explicit definitions; the project then went further and removed Fan as a logical dependency from its current fixed-order candidate proofs at $n=25,27,28,29,30$.

The audit checks the replacement as an assembled dependency chain rather than merely trusting green workflows. It verifies complete edge-range coverage of the new upper-band computations, zero final survivors in every required exact checkpoint, exact-certificate acceptance flags, the absence of the old logical Fan phrases from v2 proof surfaces, and SHA-256 integrity of every frozen historical source named by the v2 history files.

Result: **PASS** on all machine-checkable assembly checks. See `FAN_FREE_AUDIT_REPORT.json`.

The result is deliberately not described as external verification. The universal graph-to-residual lemmas and the short hand monotonicity arguments remain the main mathematical trust boundary. Historical v1 proof surfaces and Fan citations are retained rather than overwritten.

Included source: `project/reviews/n30/2026-09-09-candidate-v1/ASSEMBLY_AUDIT.md`

n=30 complete-candidate assembly audit

9 September 2026. Hostile same-assistant audit by ChatGPT/Geeps at Paul Lenz's direction.

Verdict: no blocking defect found in the assembled n=30 candidate. Independent expert review remains OPEN. This audit is not external validation.

1. Object audited

Candidate statement:

Every 30-vertex simple diameter-two edge-critical graph G satisfies $e(G) \leq 225$,
with equality exactly $K(15,15)$.

The assembly proof is `PROOF.md` in this directory.

The audit was performed against the current parameterized graph-to-demand bridge, the n=30 Delta=16 parameterization audit and clean exact workflows, the n=30 Delta=17 early-kernel package and clean replay, and a fresh standard-library outer arithmetic checker.

2. Outer edge-count reduction

The Fan expression used by the project evaluates exactly to

$$7247/32 = 226.46875$$

at n=30. Because edge count is integral and the bound is strict, an upper-bound counterexample to 225 edges can only have 226 edges.

No assumption is made that deleting an edge from a 226-edge graph leaves an edge-critical 225-edge graph. Equality at 225 is treated independently.

Verdict: no defect found.

3. Bipartite branch

A bipartite graph of diameter at most two must be complete bipartite: any missing cross-part pair has odd distance at least three. Therefore a bipartite 30-vertex graph in the class has at most $15 \cdot 15 = 225$ edges, with equality only $K(15,15)$.

$K(15,15)$ is diameter-two edge-critical: deleting a cross edge makes its endpoints distance three.

Verdict: no defect found.

4. Degree coverage

At $m=226$:

$$2m=452 > 30 \cdot 15,$$

so $\Delta \geq 16$.

At $m=225$:

$$2m=450 = 30 \cdot 15,$$

so $\Delta \geq 15$, and $\Delta = 15$ forces 15-regularity.

The assembly covers every integer Δ from the entry point through 29:

m=226: 16, 17, 18...28, 29;
m=225: 15, 16, 17, 18...28, 29.

Verdict: no missing degree sector.

5. $\Delta = 15$ equality branch

This branch exists only at $m = 225$ and is 15-regular.

For the non-bipartite branch, the cited dominating-edge theorem excludes a dominating edge at this density. Every critical edge supplies a direct or two-step witness. In either witness type, absence of a dominating edge gives degree sum at most $n - 1 = 29$. A 15-regular graph gives degree sum 30 for every pair, contradiction.

The argument does not apply the dominating-edge theorem to the bipartite equality graph; the bipartite branch is handled separately.

Verdict: no defect found. External reviewers should still check the precise hypotheses of the cited dominating-edge theorem.

6. Universal graph-to-demand bridge

For $\Delta = 16$ and 17 the assembled proof uses the same symbolic bridge:

- complement/quasi-edge construction;
- selected-edge injection;
- exact ledger $e(F) = r + t$;
- demand $S \geq r + 2t$;
- residual activity for $t > 0$;
- selected-source implication $s_i \leq \rho_u$;
- charging;
- threshold capacity.

All four $n = 30$ dense $\Delta = 16/17$ scopes have positive t :

$\Delta = 16$: $t = 2, 1$;
 $\Delta = 17$: $t = 5, 4$.

Thus residual activity is used only inside its stated $t > 0$ range.

Verdict: no new $n = 30$ bridge assumption found. The universal bridge remains the principal mathematical trust boundary.

7. $\Delta = 17$ audit

The $\Delta = 17$ route uses only the early kernel; it does not use residual-row enumeration or the cumulative-threshold endpoint LP.

Exact complete domains:

m=226: 250 charging-feasible profiles; all 250 threshold-rejected.
m=225: 1,155 charging-feasible profiles;
1,070 threshold-rejected;
67 source-count rejected;
18 exact Hall-dual rejected.

The clean workflow run 34292054922 is green. It regenerated the complete domain, recovered exactly the 18 post-threshold profiles, verified freshly generated exact duals, and independently verified the committed 18 exact integer duals using a standard-library checker.

Earlier failed CI attempts are preserved in `CI_PROVENANCE_NOTE.md`. They exposed replay-contract and stale-certificate-metadata defects; they did not produce a mathematical survivor. The committed certificate file was regenerated from the exact kernel and the subsequent run passed.

Verdict: no blocking Delta=17 defect found.

8. Parameteric isolated-C lemma

The Delta=16 row bounds require $\delta(C) \geq 1$. The expanded parameteric proof is preserved at

`project/research/n30/2026-09-08-minimal-kernel-recon-v1/ISOLATED_C_PARAMETERIC`

The key necessary consequence is

if C has an isolated vertex, then $b \leq a-1-t$.

At $(a, b) = (13, 16)$ this becomes $16 \leq 10$ at $t=2$ or $16 \leq 11$ at $t=1$, impossible. Hence $\delta(C) \geq 1$, so $d_i \leq 11$, $e(C) \geq 7$, and $r \leq 69/70$ in the two scopes.

The expanded proof explicitly checks why a quasi-edge for a missing pair inside $A \setminus \{x\}$ cannot use v or an auxiliary in A , and why the resulting residual families are disjoint.

Verdict: no defect found after hostile rederivation. This remains a high-priority external-review lemma.

9. Delta=16 finite chain

The parameterization audit records every changed constant from $n=29$ to $n=30$ and finds no hidden $n=29$ dimension assumption.

Preparation / residual rows

Clean run 34286806474:

`m=226: 48,046 charging profiles -> 2,590 demands;
50,690,620 raw residual states -> 35,530 row survivors.`

`m=225: 67,050 charging profiles -> 5,379 demands;
158,314,695 raw residual states -> 150,896 row survivors.`

The row scanner uses only necessary prefix Hall bounds and monotone supplement-cap refinement.

Exact row threshold

Clean run 34287440190 applies the exact threshold theorem using each concrete residual row and leaves

`m=226: 9 rows;
m=225: 272 rows.`

Final exact model

The $n=30$ cumulative-threshold/source-flow model is a fresh file specialized to $(a, b) = (13, 16)$ and uses the corrected per-label normalization

`sum_k n_k Z = sum_h T_h.`

Its same-group pair capacity is weakened rather than strengthened, so the relaxation may admit false survivors but cannot exclude a graph on that basis.

Clean final run 34287739057 gives

```
m=226: 9/9 exact Farkas rejections;  
m=225: 272/272 exact Farkas rejections;  
final survivors: 0 in both scopes.
```

Floating-point infeasibility is not accepted as proof; every counted exclusion has an integer Farkas certificate directly verified against the reconstructed model.

Verdict: no blocking Delta=16 defect found.

10. Delta=18 through 28

The standard-library `check_outer.py` completely enumerates sorted integer demand profiles satisfying the charging inequality for each degree and both dense edge counts.

For every Delta=18,...,28 at m=225 and m=226, the charging-feasible domain is empty.

This is stronger than the earlier scalar h-index reconnaissance and removes any need to inspect these degree values separately.

Verdict: no survivor and no missing case found.

11. Delta=29

A universal vertex forces a star in a diameter-two edge-critical graph: any edge among the other vertices could otherwise be deleted without increasing diameter beyond two. The star has 29 edges.

Verdict: excluded.

12. Equality assembly

At 225 edges:

- bipartite equality gives K(15,15);
- non-bipartite Delta=15 is impossible;
- Delta=16 and 17 are computationally/exactly excluded;
- Delta>=18 is charging-excluded or star.

No other equality branch remains.

Verdict: equality uniqueness assembled correctly.

13. Findings from the hostile process

No blocking mathematical defect was found, but the process did find and preserve several nontrivial audit issues:

1. the isolated-C proof needed a more explicit parameteric derivation;
2. requiring freshly proposed LP dual rays to reproduce the same certificate bytes/vectors was an invalid replay invariant;
3. one committed Delta=17 dual record had stale derived `lhs/rhs` metadata despite valid weights; the exact checker caught it and the file was regenerated before the clean successful replay.

These findings increase confidence in the audit process but do not constitute external independence.

14. Final audit verdict

Within the current audited bridge framework, all $n=30$ dense sectors close and equality is uniquely $K(15,15)$.

Project status recommended: COMPLETE CANDIDATE, not theorem.

Promotion beyond candidate requires at minimum independent expert review of the universal graph lemmas and independent scrutiny/reproduction of the computational relaxations.